

SOFTWARE ENGINEERING LAB 2

Agile Backlog Creation & Sprint Simulation in Jira

Digital Library Management System

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1. Objective

The objective of this lab is to use Jira to convert functional requirements into Agile backlog items, organize them into Epics and User Stories, prioritize and estimate the stories using Story Points, simulate two Scrum sprints, and analyse sprint progress using Jira.

2. Problem Scenario

The selected scenario for this lab is a Digital Library Management System. The Agile backlog was created from the functional requirements identified previously. The backlog was organized into major functional themes (Epics), with related User Stories representing end-user goals.

3. Epics

The functional requirements were grouped into the following major Epics:

Epic	Epic Name	Purpose / Description
DL-1	Library Catalog Search	Supports searching and locating library resources.
DL-2	Book Reservation Management	Supports book holds, reservation queue management, and availability notification.
DL-3	Fine Management	Supports viewing overdue fine information.

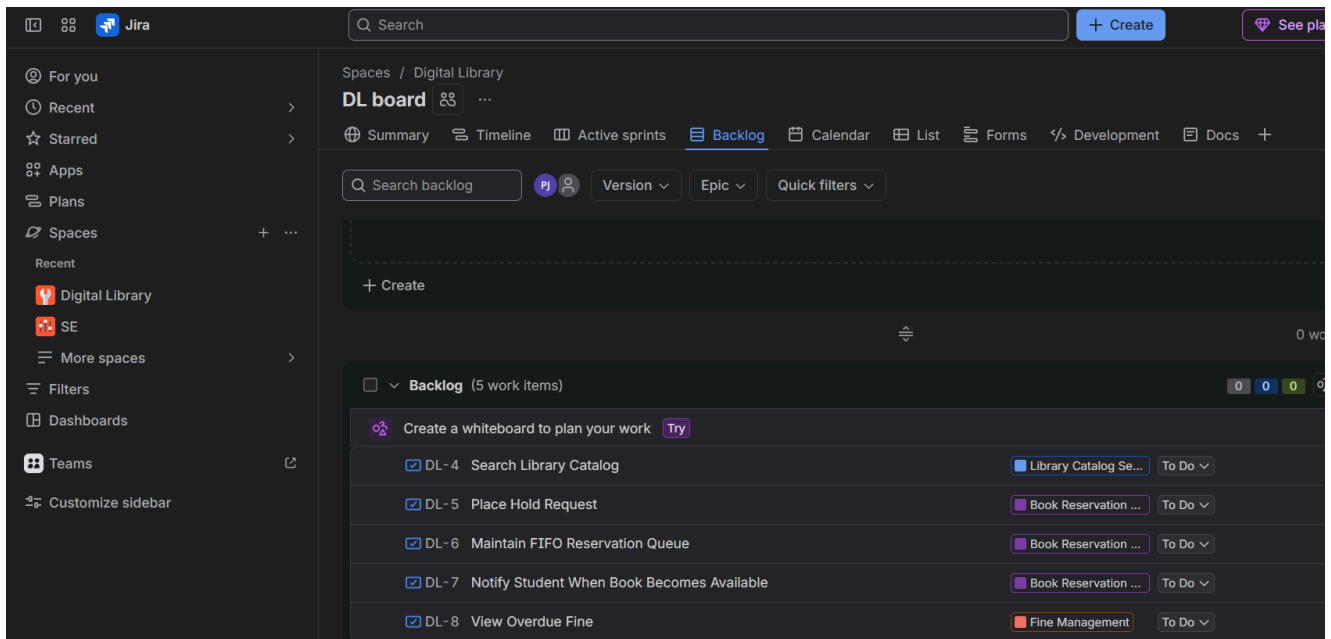


Figure: BACKLOG / EPICS VIEW

4. User Stories

The following User Stories were created using the Agile format: “As a [user role], I want [goal], so that [benefit].”

Key	User Story	Epic	Priority	Story Points
DL-4	Search Library Catalog	Library Catalog Search	High	3
DL-5	Place Hold Request	Book Reservation Management	High	5
DL-6	Maintain FIFO Reservation Queue	Book Reservation Management	High	5
DL-7	Notify Student When Book Becomes Available	Book Reservation Management	High	3
DL-8	View Overdue Fine	Fine Management	Medium	3

4.1 User Story Details

DL-4 – Search Library Catalog

As a student member, I want to search the physical library catalog using an ISBN or book title, so that I can quickly locate the required library resources.

Acceptance Criteria: The student can search using an ISBN or book title; for a valid query, the system displays matching catalog records.

DL-5 – Place Hold Request

As a student member, I want to place a hold request on a book, so that I can reserve it when it is unavailable.

Acceptance criteria should reflect successful creation of a hold request and the reservation being recorded.

DL-6 – Maintain FIFO Reservation Queue

As a library system/user, I want reservation requests to be maintained in FIFO order, so that books are allocated fairly according to request order.

Acceptance criteria should reflect that reservation requests are processed in first-in-first-out order.

DL-7 – Notify Student When Book Becomes Available

As a student member, I want to be notified when a reserved book becomes available, so that I know when I can collect or borrow it.

Acceptance criteria should reflect notification when the reserved book becomes available.

DL-8 – View Overdue Fine

As a student member, I want to view my overdue fine, so that I can know the amount that needs to be paid.

Acceptance criteria should reflect display of the student's applicable overdue fine information.

5. Backlog Prioritization and Story Points

The User Stories were prioritized according to their importance to the Digital Library workflow. Story Points were assigned using the Fibonacci-style scale specified in the lab handout. Story Points represent relative effort, complexity, and uncertainty rather than hours.

The assigned Story Points were: 3, 5, 5, 3, and 3, giving a total backlog estimate of 19 Story Points.

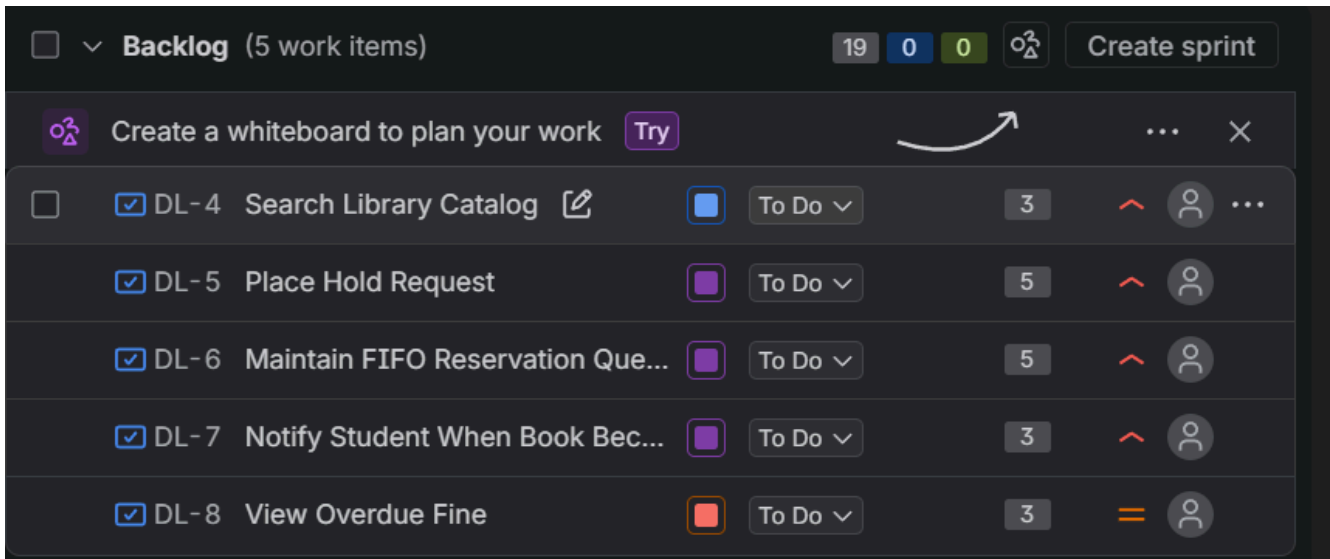


Figure: STORY POINT ASSIGNMENTS

6. Sprint 1

Sprint 1 was created as the first one-week sprint. Three stories were selected based on priority and planned scope.

Story	Key	Story Points	Sprint Result
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Search Library Catalog	DL-4	3	Done
Place Hold Request	DL-5	5	Done
Maintain FIFO Reservation Queue	DL-6	5	Done

Total Sprint 1 estimate: 13 Story Points.

Sprint Goal: Complete core catalog search and book reservation management functionality.

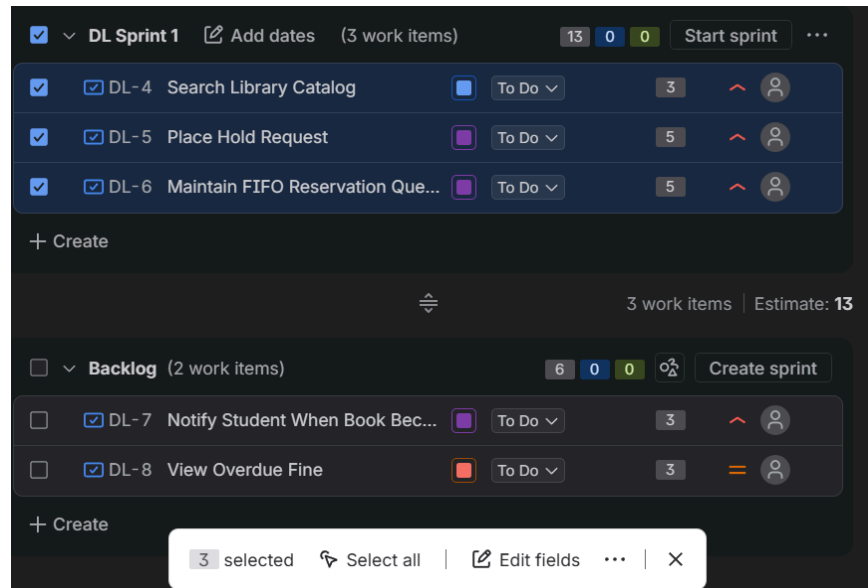


Figure: ACTIVE SPRINT BOARD

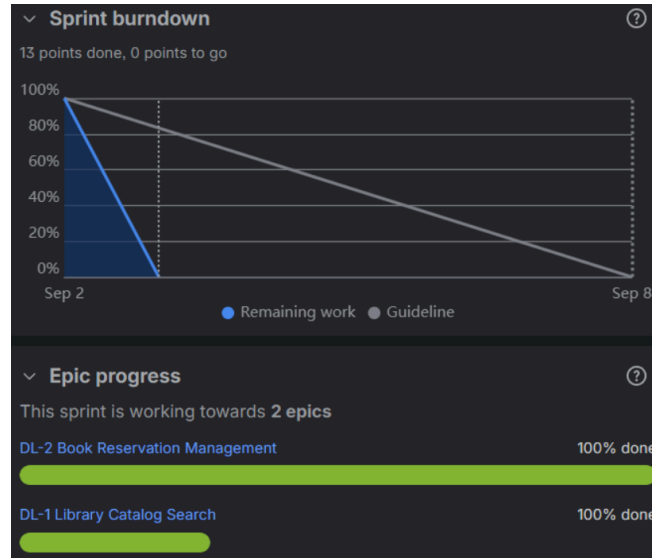


Figure: SPRINT 1 – BURNDOWN CHART

7. Sprint 2

Sprint 2 was created after Sprint 1 and contained the remaining two User Stories.

Story	Key	Story Points	Sprint Result
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Notify Student When Book Becomes Available	DL-7	3	Done
View Overdue Fine	DL-8	3	Done

Total Sprint 2 estimate: 6 Story Points.

Sprint Goal: Complete book availability notification and overdue fine management functionality.

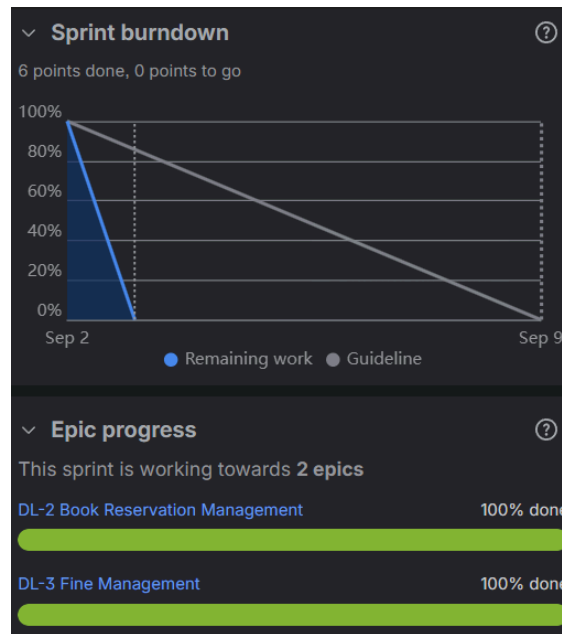


Figure: SPRINT 2 – BURNDOWN CHART

8. Sprint and Backlog Summary

Item	Stories	Story Points	Status
Sprint 1	3	13	Completed
Sprint 2	2	6	Completed
Overall	5	19	All planned stories completed

Both simulated sprints were completed successfully. The full backlog consisted of 5 User Stories and 19 Story Points, and all planned stories reached the Done state.

9. Burndown Chart Analysis

The Jira burndown charts were used to observe remaining work against the guideline for each sprint. Sprint 1 finished with 13 Story Points completed and 0 points remaining. Sprint 2 finished with 6 Story Points completed and 0 points remaining.

The charts show a rapid reduction in remaining work in the simulated sprints. Because this was a simulation rather than a real development cycle, the sharp drop should be interpreted as evidence of the simulated completion pattern rather than as a precise measurement of real team velocity.

10. Reflection

1. Did your estimations reflect the actual effort?

Yes. The Story Points provided a reasonable relative estimate of effort and complexity. More involved reservation-management stories were assigned 5 points, while comparatively smaller stories were assigned 3 points. All planned stories were completed in the simulated sprints, indicating that the estimates were reasonable for this exercise.

2. Was your backlog well-prioritized?

Yes. The backlog was prioritized according to the importance of the functionality to library users. Core catalog and reservation functionality was handled in the first sprint, while notification and fine management functionality was handled in the second sprint.

3. How did your simulated sprint align with your plan?

The simulated sprints aligned with the planned scope. Sprint 1 contained three stories totaling 13 Story Points, and Sprint 2 contained two stories totaling 6 Story Points. All planned stories were moved through the workflow and completed.

4. What insights did the burndown chart give about your team's capacity?

The burndown charts showed that the selected workload could be completed within the simulated sprint periods. Both sprints reached zero remaining work. The rapid decrease in remaining work also highlights that the exercise was a simulation and that real project velocity would require data from actual development work over multiple sprints.

11. Conclusion

This lab demonstrated the use of Jira for Agile backlog management and Scrum sprint simulation. The Digital Library requirements were converted into Epics and User Stories, prioritized, estimated using Story Points, and organized into two one-week sprints. Both sprints were completed successfully, and the burndown charts were used to analyse progress and remaining work.